Emanuele Sansone

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Research Interests and Vision: His research interests lie at the intersection between unsupervised learning and mathematical logic. His research vision is to make machines curious and creative by empowering them with the capability to acquire and discover knowledge from data in an autonomous, scalable and reliable manner.

Employment

2024-now **Marie-Curie Global Fellow, MIT CSAIL & KU Leuven ESAT**
Topic: Unsupervised Statistical Relational Learning

2023-2024 **Post-Doctoral Researcher, KU Leuven ESAT**
Topic: Unsupervised Deep Learning

2020-2023 **Post-Doctoral Researcher, KU Leuven CS**
Topic: Neuro-Symbolic Learning

2018-2020 **Research Scientist, Huawei R&D UK Ltd**
Topic: Unsupervised Deep Learning

2015-2016 **Visiting Researcher, Nanjing University CS**

Education

2013-2018 **PhD, Inform. Engin. and Computer Science, Unversity of Trento**
Advisor: Francesco G. B. De Natale
Thesis : "Towards Uncovering the True Use of Unlabeled Data in Machine Learning"

2011-2012 **Master degree, Telecommunications Engineering, University of Trento**
Advisor: N. Conci, G. Boato
Thesis: "Multimodal Photo Galleries Synchronization"
Award: 110/110 cum laude

2008-2010 **Bachelor degree, Telecommunications Engineering, University of Trento**
Advisor: D. Petri, M. Corrà
Thesis: "Master P-NET Protocol Implementation on PIC 32 Architecture"

Grants and Awards

2024-now **MSCA Postdoctoral Global Fellowship Grant (177.322 €)**
Project: "Discovering the World Through Unsupervised Statistical Relational Learning"

2025-2026 **LUMI Compute Grant Award (162.737 €)**
Project: "Self-Supervised Learning of Composable Concept Hierarchies".

2024-2025 **FWO Compute Grant Award (32.963 €)**
Project: "Scaling Failure-Free Representation Learning"

2021-2023 **Outstanding Reviewer Award**
ICLR 2021, ICML 2022, NeurIPS 2022, AISTATS 2023, NeurIPS 2023

Publication Overview

Published 3 journal, 8 conference and 9 workshop papers (all peer-reviewed) in venues such as ICML, NeurIPS, ICLR, ECAI, TPAMI, TMLR and CogSci. 2 papers are **single author** and 4 papers are **last author**.

Under review	**E. Sansone**, A. Solar-Lezama (2026). Power Term Polynomial Algebra for Boolean Logic
TMLR	**E. Sansone**, R. Manhaeve (2025). Unifying Self-Supervised Clustering and Energy-Based Models. *Transactions of Machine Learning Research*
ICML	**E. Sansone**, T. Lebailly, T. Tuytelaars (2025). Collapse-Proof Non-Contrastive Self-Supervised Learning. *International Conference on Machine Learning*
ECAI	V. Verreet, L. De Smet, L. De Raedt, **E. Sansone** (2024). EXPLAIN, AGREE, LEARN: Scaling Learning for Neural Probabilistic Logic. *European Conference on Artificial Intelligence*
NeurIPS	L. De Smet, **E. Sansone**, P. Z. D. Martires (2023). Differentiable Sampling of Categorical Distributions Using the CatLog-Derivative Trick. *Neural Information Processing Systems*
NeurIPS	E. Misino, G. Marra, **E. Sansone** (2022). VAEL: Bridging Variational Autoencoders and Probabilistic Logic Programming. *Neural Information Processing Systems*
ICML	**E. Sansone** (2022). LSB: Local Self-Balancing MCMC in Discrete Spaces. *International Conference on Machine Learning*
TPAMI	**E. Sansone**, F.G.B De Natale, Z.H. Zhou (2018). Efficient Training for Positive Unlabeled Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*

Service

Action Editor	**TMLR**: Transactions of Machine Learning Research (2024-now)
Guest Editor	**Machine Learning (Springer)**: Special Issue on Learning and Reasoning (2024-now)
Area Chair	**IJCLR**: International Joint Conference on Learning & Reasoning (2024-now) **ICLR**: International Conference on Learning Representations (2026) **NeurIPS**: Neural Information Processing Systems (2026)
Organizer	**Hybrid micro-workshop**: “Synergies among Neuro-Symbolic, Graph Embeddings and Language Models”, KU Leuven Attendees/speakers from KU Leuven, TU Darmstadt, EPFL, Fraunhofer IAIS and University of Siena.
Co-Organizer	**Online workshop**: “What are the Next Measurable Challenges in AI?” **Online workshop**: “Learning and Reasoning” Invited talk from Richard Evans (DeepMind) **Online workshop**: “Unifying AI Paradigms and Representations”
Reviewer	**TPAMI**: IEEE Transactions on Pattern Analysis and Machine Intelligence (2022-now) **TNNLS**: IEEE Transactions on Neural Networks and Learning Systems (2022-now) **JMLR**: Journal of Machine Learning Research (2020-now) **Machine Learning (Springer)** (2020-now) **IJAR**: International Journal of Approximate Reasoning (2022-now) **TIP**: IEEE Transactions on Image Processing (2023-now)
PC Reviewer	**ICLR**: International Conference on Learning Representations (2021-2025) **NeurIPS**: Neural Information Processing Systems (2021-2025)

ICML: International Conference on Machine Learning (2022-2025)
AISTATS: Artificial Intelligence and Statistics (2022-2024)
CogSci: Cognitive Science (2024-2025)
CVPR: Computer Vision and Pattern Recognition (2024-2025)

Teaching, Mentorship in Academia

2022 **Teaching Assistant, Uncertainty in Artificial Intelligence** (B-KUL-H00H2a), *KU Leuven*

2020-2022 **Teaching Assistant, Machine Learning and Inductive Inference** (B-KUL-H00G6a), *KU Leuven*

2016 **Teaching Assistant, Computer Vision** (Code 140266), *University of Trento*

2015 **Teaching Assistant, Multimedia Networking** (Code 140151), *University of Trento*

2021-now **Co-supervision of PhD Students**
2025-now Leonardo Hernandez Cano: "Library Learning for Puzzle Solving", *MIT*
2025-now Nikola Dukic: "Compositional Self-Supervised Learning", *KU Leuven*
2024-2025 Tim Lebailly: "Collapse-Proof Self-Supervised Learning", *KU Leuven*
2024-2025 Lennert De Smet: "Scalable Neurosymbolic Learning", *KU Leuven*
2024 Victor Verreet: "Probabilistic Neurosymbolic Learning", *KU Leuven*
2024-now Vincenzo Collura: "Benchmarking Neurosymbolic Generation", *University of Luxembourg*
2022-2025 Eleonora Misino: "Integrating Domain Knowledge in Data-Driven AI Approaches", *University of Bologna*

2020-now **Supervision of Master Theses**
2022-2023 Vincenzo Collura: "Neurosymbolic Learning: Challenges and Benchmarks", *University of Bologna*
2021-2022 Rik Bossuyt: "Predicting SAT with Graph Neural Networks", *KU Leuven*
2020-2021 Eleonora Misino: "Deep Generative Models with Probabilistic Logic Priors", *University of Bologna*

2021-now **Supervision of Honor/Undergraduate Students**
2025-now Yifei Jin, Naaisha Agarwal, Nishad Mansoor: "Neuro-Symbolic Generative AI Challenge", *MIT*
2025-now Andrew Zhang: "What is the Similarity Metric in Diffusion Models?", *MIT*
2022-2023 Felix Huyghe, Sander Schildermans: "Learning to Predict the Ball Trajectory in Foosball Tables", *KU Leuven*
2021-2022 Xander Haijen: "Comparison of Reinforcement Learning Methods Applied to Tetris", *KU Leuven*

2018-2020 **Industry Supervision of Intern Students** at Huawei
Yinbai Li (Bachelor student in Mathematics at *University of Cambridge*), Xingyu Jin (Master student in Computer Science at *University of Edinburgh*)

Teaching, Mentorship in Sports

2011-now **Ski Master Instructor**
Certificate released by *Federazione Italiana Sport Invernali* (FISI), Milan (Italy).

Description: Highest professional degree in alpine skiing in Italy, which can be achieved by demonstrating excellent skiing capabilities as well as strong organizational, didactical and methodological skills used when teaching/coaching. (The total number of ski master instructors in Italy is around 200 (scroll "Sci Alpino" in "Disciplina" and check "Istruttori").
Responsibilities: Holding several professional education and training courses for ski instructors and candidate ski instructors. People prepared by me to become ski instructors (all of them are now coaches and/or ski instructors): Alessandro Berlanda, Camilla Berlanda, Martina Kerschbaumer, Davide Raineri, Valentina Zampedri, Erman Baldessari, Federico Tonezzer, Martina Longobardi, Chiara Villotti, Stefano Gonzo, Francesca Cella, Samuel Piffer, Teo Valle, Martino Santoni, Emma Santoni, Thomas Corradino, Silvia Zeni, Marco Faccenda.

2007-2017 **Ski Coach (3° level)**
Certificate released by *Federazione Italiana Sport Invernali* (FISI), Milan (Italy).
Description: Professional degree for coach racing teams at local, regional and national level.
Responsibilities: Coach of young athletes (6-20 years old) in several racing ski teams: Ski Team Sopramonte (2007/2008), Sci Club Padova (2008/2009), Sci Club Panarotta (2009-2011), Ski Team Paganella (2013/2014), Campiglio Ski Team (2016/2017).

2007-2017 **Ski Instructor**
Certificate released by Autonomous Province of Trento, Trento (Italy).

Talks

Inv. Talk
- "How Can We Make Neural Networks Both Unsupervised and Trustworthy?". *Boston AI/ML User Group Meetup, online* (2025)
- "From Neuro-Symbolic Learning to Unsupervised Deep Induction". *Department of Computer Science, TU Graz* (2025)
- "What's Past & What's Next: Learning & Learning to Acquire Knowledge". *Department of Computer Science, University of Manchester* (2023)
- "Discovering the World Through Unsupervised Statistical Relational Learning". *School of Informatics, University of Edinburgh* (2023)
- "Coulomb Autoencoders". Symposium on Generative Networks in Computer Vision and Machine Learning, *British Machine Vision Association, London* (2019)

Seminar
- "Collapse-Proof Non-Contrastive Self-Supervised Learning". *Deparment of Electrical Engineering, KU Leuven* (2026)
- "Unifying Self-Supervised Clustering and Energy-Based Models". *Computer Science & Artificial Intelligence Laboratory, MIT* (2025)
- "Collapse-Proof Non-Contrastive Self-Supervised Learning". *Computer Science & Artificial Intelligence Laboratory, MIT* (2025)
- "Self-Supervised ... Generative Learning: XOR, AND or IFF?". *Department of Mathematics and Statistics, South Dakota State University* (2023)
- "GEDI: GEnerative and DIscriminative Training for Self-Supervised Learning". *Department of Computer Science, KU Leuven* (2023)
- "Generative Adversarial Networks". *Fondazione Bruno Kessler, Trento* (2017)

Presentat.
- "Foundations of Unsupervised Neural Induction". *Graduate School of Education, Stanford University* (2025)
- "Collapse-Proof Non-Contrastive Self-Supervised Learning". *Computer Science & Artificial Intelligence Laboratory, MIT* (2025)
- "Collapse-Proof Non-Contrastive Self-Supervised Learning". *Faculty of Arts and Sciences, Harvard University* (2025)
- "Discovering the World Through Unsupervised Statistical Relational Learning". *Inter-Departmental Postdoc Symposium, MIT* (2025)

- “GEDI: GEnerative and DIscriminative Training for Self-Supervised Learning”. *Department of General and Computational Linguistics, University of Tübingen* (2023)
- “LSB: Local Self-Balancing MCMC in Discrete Spaces”. *ICML, Baltimore* (2022)
- “Coulomb Autoencoders”. *ECAI, online* (2020)
- “Classtering: Joint Classification and Clustering with Mixture of Factor Analysers”. *ECAI, The Hague* (2016)

Manuscripts Under Review

Journ. **E. Sansone**, A. Solar-Lezama (2026). Power Term Polynomial Algebra for Boolean Logic

Conf. L. Hernandez Cano, I. Zareski, L. El Amouri, P. Zhao, **E. Sansone**, M. Mascini, Y. Pu, B. Zhao, M. Kryven (2026). Prospective Compression in Human Abstraction Learning

Conf. P. Zhao, L. Hernandez Cano, **E. Sansone**, B. Zhao, M. Kryven (2026). Adaptive Online Learning of Structured Abstractions in Visual Problem-Solving

Journal Publications

TMLR **E. Sansone**, R. Manhaeve (2025). Unifying Self-Supervised Clustering and Energy-Based Models. *Transactions of Machine Learning Research*

TPAMI **E. Sansone**, F.G.B De Natale, Z.H. Zhou (2018). Efficient Training for Positive Unlabeled Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*

TMM **E. Sansone**, K. Apostolidis, N. Conci, G. Boato, V. Mezaris, F. G. B. De Natale (2017). Automatic Synchronization of Multi-User Photo Galleries. *IEEE Transactions on Multimedia*

Conference Publications

CogSci P. Zhao, **E. Sansone**, M. Kryven, B. Zhao (2026). Online Library Learning in Human Visual Puzzle Solving. *Annual Meeting of the Cognitive Science Society*

ICML **E. Sansone**, T. Lebailly, T. Tuytelaars (2025). Collapse-Proof Non-Contrastive Self-Supervised Learning. *International Conference on Machine Learning*

ECAI V. Verreet, L. De Smet, L. De Raedt, **E. Sansone** (2024). EXPLAIN, AGREE, LEARN: Scaling Learning for Neural Probabilistic Logic. *European Conference on Artificial Intelligence*

NeurIPS L. De Smet, **E. Sansone**, P. Z. D. Martires (2023). Differentiable Sampling of Categorical Distributions Using the CatLog-Derivative Trick. *Neural Information Processing Systems*

NeurIPS E. Misino, G. Marra, **E. Sansone** (2022). VAEL: Bridging Variational Autoencoders and Probabilistic Logic Programming. *Neural Information Processing Systems*

ICML **E. Sansone** (2022). LSB: Local Self-Balancing MCMC in Discrete Spaces. *International Conference on Machine Learning*

ECAI **E. Sansone**, H. T. Ali, J. Sun (2020). Coulomb Autoencoders. *European Conference on Artificial Intelligence*

ECAI **E. Sansone**, A. Passerini, F. G. B. De Natale (2016). Classtering: Joint Classification and Clustering with Mixture of Factor Analysers. *European Conference on Artificial Intelligence*

Peer-Reviewed Workshops and Technical Reports

CVPR	N. Agarwal, Y. Wu, Y. Jiang, Y. Hu, N. Mansoor, M. Li, Y. Peng, W.Z. Dai, Y.X. Ding, **E. Sansone** (2026). OrigamiBench: An Interactive Environment to Synthesize Flat-Foldable Origamis. *CVPR Workshop MAR* - **Oral**
ICML	B. Kim, M. Puthawala, J. Chul Ye, **E. Sansone** (2024). (Deep) Generative Geodesics. *ICML Workshop GRaM*
NeurIPS	**E. Sansone** (2023). The Triad of Failure Modes and a Possible Way Out. *NeurIPS Workshop SSLTheoryPractice*
ICML	L. De Smet, **E. Sansone**, P. Z. D. Martires (2023). Differentiable Sampling of Categorical Distributions Using the CatLog-Derivative Trick. *ICML Workshop DiffAE*
ICML	V. Verreet, L. De Smet, **E. Sansone** (2023). EXPLAIN, AGREE and LEARN: A Recipe for Scalable Neurosymbolic Learning. *ICML Workshop KLR*
NeSy	E. Misino, G. Marra, **E. Sansone** (2023). VAEL: Bridging Variational Autoencoders and Probabilistic Logic Programming (Extended Abstract). *NeSy Workshop*
ICLR	E. Misino, G. Marra, **E. Sansone** (2023). VAEL: Bridging Variational Autoencoders and Probabilistic Logic Programming. *ICLR Workshop NeSy-GeMs*
IJCAI	**E. Sansone**, R. Manhaeve (2023). Learning Symbolic Representations Through Joint GEnerative and DIscriminative Training (Extended Abstract). *IJCAI Workshop KBCG*
ICLR	**E. Sansone**, R. Manhaeve (2023). Learning Symbolic Representations Through Joint GEnerative and DIscriminative Training. *ICLR Workshop NeSy-GeMs*
Report	**E. Sansone**, R. Manhaeve (2022). GEDI: GEnerative and DIscriminative Training for Self-Supervised Learning
Report	**E. Sansone** (2021). Leveraging Hidden Structure in Self-Supervised Learning. *Technical Report*
Thesis	**E. Sansone** (2018). Towards Uncovering the True Use of Unlabeled Data in ML. *PhD Thesis*